EXPRIMENT-12

12.) Hill Cipher Encryption and Decryption

Aim:

To write a Python program to encrypt and decrypt a message using the Hill cipher.

PROGRAM:

```
def encrypt(text):
    text = ''.join(ch for ch in text.lower()
                   if ch.isalpha())

    if len(text) % 2 != 0:
        text += 'x'

    result = ""

    for i in range(0, len(text), 2):
        x = ord(text[i]) - 97
        y = ord(text[i+1]) - 97

        a = (9*x + 4*y) % 26
        b = (5*x + 7*y) % 26

        result += chr(a + 97)
        result += chr(b + 97)

    return result
```

```
    result = ""

    for i in range(0, len(text), 2):
        x = ord(text[i]) - 97
        y = ord(text[i+1]) - 97

        a = (23*(7*x - 4*y)) % 26
        b = (23*(-5*x + 9*y)) % 26

        result += chr(a + 97)
        result += chr(b + 97)

    return result


message = "meet me at the usual place at ten rather than eight oclock"

cipher = encrypt(message)

print("Plain Text:", message)
print("Cipher Text:", cipher)
print("Decrypted Text:", decrypt(cipher))
```

OUTPUT:

```
Plain Text: meet me at the usual place at ten rather than eight oclock
Cipher Text: ukixukydromeiwszxwiokunukhxhroajroanqyebtlkjegad
Decrypted Text: meetmeattheusualplaceattenratherthaneightoclockx
```

RESULT:

Thus, the message was successfully encrypted and decrypted using the Hill cipher.